

Introduction to Machine Learning

Machine learning is a branch of artificial intelligence that enables systems to learn from data and improve their performance over time.

Supervised learning algorithms learn a mapping function from input variables to an output variable using labelled training data.

Unsupervised learning discovers hidden structure in unlabelled data. Clustering algorithms such as k-means and hierarchical clustering are common examples.

Reinforcement learning trains an agent to make sequential decisions by rewarding desired behaviours and penalizing undesired ones.

Neural Networks and Deep Learning

Artificial neural networks are computational models loosely inspired by the biological neural networks in the brain.

Deep learning refers to neural networks with multiple hidden layers. Convolutional neural networks (CNNs) excel

Transformer architectures, introduced by Vaswani et al. in 2017, use self-attention mechanisms to relate position